

Applied Statistics (MATH10096)

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Linear models in general

- ↪ The simple linear model we used in the Hubble's law example can be generalized by allowing the response variable to depend on multiple predictor variables.
- ↪ These extra predictor variables can themselves be transformations of the original predictors.
- ↪ Examples of linear models:
 - $\mu_i = \beta_1 + \beta_2 x_i, \quad Y_i = \mu_i + \epsilon_i.$
 - $\mu_i = \beta_1 + \beta_2 x_i + \beta_3 x_i^2 + \beta_4 x_i^3, \quad Y_i = \mu_i + \epsilon_i.$
 - $\mu_i = \beta_1 + \beta_2 x_{i1} + \beta_3 x_{i2} + \beta_4 \log(x_{i1} x_{i2}), \quad Y_i = \mu_i + \epsilon_i.$
- ↪ Note that the parameters need to enter the model in a linear way but the predictor variables can enter the model nonlinearly.

Linear models in general

↪ The following model is **not linear in the parameters**:

$$Y_i = \beta_1 + \beta_2 x_i^{\beta_3} + \epsilon_i.$$

↪ Some relationships can be transformed to linearity!

↪ For example, the following model can be converted into a linear model by taking logarithms.

$$Y_i = \beta_1 x_i^{\beta_2} \epsilon_i.$$

Matrix notation

↪ A linear model can generally be written as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where

- ↪ $\mathbf{y}$ is the **response variable** vector and has dimension $n \times 1$.
- ↪ $\mathbf{X}$ is the **model matrix**, which has dimension $n \times p$, where p denotes the number of parameters to be estimated including, in general, the intercept.
- ↪ $\boldsymbol{\beta}$ is the **parameter vector** (or **coefficient vector**) and has dimension $p \times 1$.
- ↪ $\boldsymbol{\epsilon}$ is a **random error** vector of dimension $n \times 1$ with independent components and such that $\mathbb{E}(\boldsymbol{\epsilon}) = \mathbf{0}$ and the variance-covariance matrix of $\boldsymbol{\epsilon}$ is $\mathbf{V}_{\boldsymbol{\epsilon}} = \text{var}(\boldsymbol{\epsilon}) = \sigma^2 \mathbf{I}_n$.

Matrix notation

- ↪ The **expected value** vector $\mu = \mathbb{E}(\mathbf{y})$ has dimension $n \times 1$ and according to the linear model, $\mu = \mathbf{X}\beta$.
- ↪ Once the model is estimated, the **parameter estimates** are denoted by $\hat{\beta}$. This notation will also be used for the parameter **estimator**, the context indicating which is meant.
- ↪ The **fitted value** vector is $\hat{\mu} = \mathbf{X}\hat{\beta}$.
- ↪ The **residual** vector is $\hat{\epsilon} = \mathbf{y} - \hat{\mu}$.

Linear model theory (Section 4 of the lecture notes)

- As in the simple one parameter case, point estimates of the **linear model parameters, β , can be obtained by the method of least squares**; that is, by minimising the residual sum of squares

$$S(\beta) = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - \sum_{j=1}^p \beta_j x_{ij})^2 = \sum_{i=1}^n (y_i - \mu_i)^2, \quad \mu_i = \mathbf{x}_i^T \beta.$$

- Here $x_{i1} = 1, \forall i = 1, \dots, n$, for design matrices including an intercept term and $\mathbf{x}_i^T = (x_{i1}, \dots, x_{ip})$ is the i th row of the design matrix, $\mathbf{X}$
- From now on the random vector, $\mathbf{y}$, and its observed value will not be distinguished notationally, the context making clear which is meant.
- Recall that if $\mathbf{v}$ is any vector of dimension, $n \times$, then

$$\|\mathbf{v}\|^2 \equiv \mathbf{v}^T \mathbf{v} \equiv \sum_{i=1}^n v_i^2.$$

Linear model theory (Section 4 of the lecture notes)

↪ Hence

$$\begin{aligned} S(\beta) &= \|\mathbf{y} - \boldsymbol{\mu}\|^2 = \|\mathbf{y} - \mathbf{X}\beta\|^2 \\ &= (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta) \\ &= (\mathbf{y}^\top - \beta^\top \mathbf{X}^\top) (\mathbf{y} - \mathbf{X}\beta) \\ &= \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}\beta - \beta^\top \mathbf{X}^\top \mathbf{y} + \beta^\top \mathbf{X}^\top \mathbf{X}\beta \\ &= \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\beta + \beta^\top \mathbf{X}^\top \mathbf{X}\beta \end{aligned}$$

↪ Differentiating $S(\beta)$ with respect to β and setting the corresponding system of equations to zero leads to

$$\frac{\partial S(\beta)}{\partial \beta} = \mathbf{0} \Rightarrow -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\beta = \mathbf{0} \Rightarrow \mathbf{X}^\top \mathbf{X}\beta = \mathbf{X}^\top \mathbf{y}.$$

↪ If the design matrix $\mathbf{X}$ is full rank, i.e., $\text{rank}(\mathbf{X}) = p$, so that $\mathbf{X}^\top \mathbf{X}$ is also full rank and therefore invertible, we then have

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Linear model theory (Section 4 of the lecture notes)

↪ It can be easily shown that $\hat{\beta}$ is unbiased (has expectation equal to the true β):

$$\mathbb{E}(\hat{\beta}) = \mathbb{E}[(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}] = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} \beta = \beta.$$

↪ The variance-covariance matrix of $\hat{\beta}$ is also easy to derive:

$$\begin{aligned} \mathbf{V}_{\hat{\beta}} &= \text{var}(\hat{\beta}) = \text{var}\{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}\} \\ &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \text{var}(\mathbf{y}) (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \\ &= \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}. \end{aligned}$$

Linear model theory (Section 4 of the lecture notes)

- ↪ In numerical linear algebra, there is a 'rule' that we should avoid computing the inverse of a matrix whenever possible.
- ↪ The following example is an extreme case that *illustrates the why of this 'rule'*.

Linear model theory (Section 4 of the lecture notes)

```
# Number of observations
n <- 50
# One predictor only
x <- seq(1, 500, len = n)
# Design matrix (polynomial regression)
X <- cbind(1, x, x^2, x^3) #Thus p = 4
colnames(X) <- c("Intercept", "x", "x2", "x3")
# True parameter vector
beta <- matrix(c(1, 1, 1, 1), nrow = 4, ncol = 1)
# Fixing the seed so that results are reproducible
set.seed(1)
# Simulating the response data
y <- X%*%beta + rnorm(n)

# OLS estimate of beta
solve(t(X)%*%X)%*%t(X)%*%y

## Error in solve.default(t(X) %*% X): system is computationally singular: reciprocal
condition number = 2.93617e-17

lm_fit <- lm(y ~ X - 1)
lm_fit$coefficients

## XIntercept      Xx      Xx2      Xx3
## 0.9038373 1.0066440 0.9999622 1.0000001
```